RAG Learning

OCR B A-Level Physics — Practical Activities

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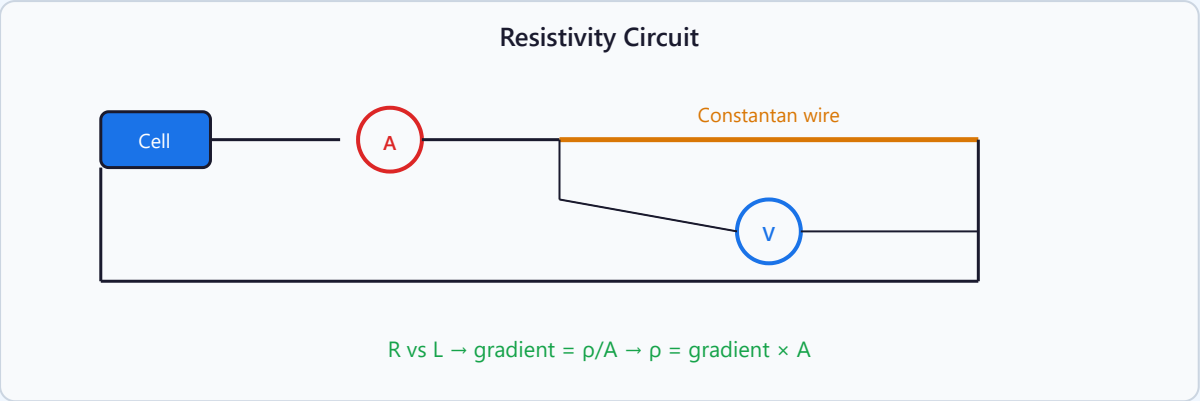
⚗ Resistivity of a Wire



Apparatus

- Constantan wire (~1 m, taped to ruler)
- Ammeter, voltmeter, cell/power supply, switch
- Micrometer screw gauge, metre ruler, crocodile clips

Method

1. Measure wire diameter at 6+ points with micrometer. Calculate $A = \pi d^2/4$.
2. Connect ammeter in series, voltmeter in parallel across wire section.
3. Use clips to select length L. Record V and I. Calculate $R = V/I$.
4. Repeat for 6+ lengths. Plot R vs L.
5. Gradient = ρ/A, so ρ = gradient × A.

Safety

- Switch off between readings; low voltage; avoid touching hot wire

Analysis

$$R = \rho L/A \Rightarrow \text{gradient of R vs L} = \rho/A$$

Expected: ρ(constantan) ≈ 4.9×10^{-7} Ω m.

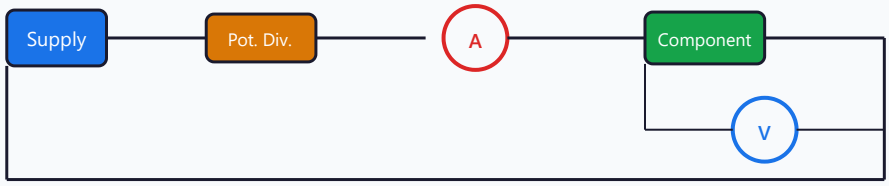
Source of Error	Type	Improvement
Wire heating	Systematic	Switch off between readings; low current

Contact resistance	Systematic	Clean clips; firm connections
Diameter variation	Random	Multiple micrometer readings; use mean
Length measurement at clip	Random	Tape wire to ruler; align carefully

Exam Tip: Constantan is chosen because its resistance barely changes with temperature. The biggest uncertainty typically comes from the diameter (squared in the area calculation). A non-zero y-intercept suggests contact resistance.

V-I Characteristics

V-I Characteristics Circuit



Components: ohmic conductor, filament lamp, diode

Use potential divider for full 0–V range

Apparatus

- DC power supply with potential divider (or variable resistor)
- Ammeter, voltmeter
- Components to test: wire (ohmic), filament lamp, silicon diode
- Switch, connecting wires

Method

1. Connect the potential divider circuit for full voltage range (0 to V_{max}).
2. Place the component under test in series with the ammeter. Voltmeter in parallel across it.

3. Vary voltage from 0 upwards. Record V and I at each setting.
4. For the diode, reverse the connections and repeat.
5. Plot I vs V for each component.

Safety

- Filament lamp gets hot — do not touch
- Low voltage (< 12 V)
- Do not exceed maximum current rating of components

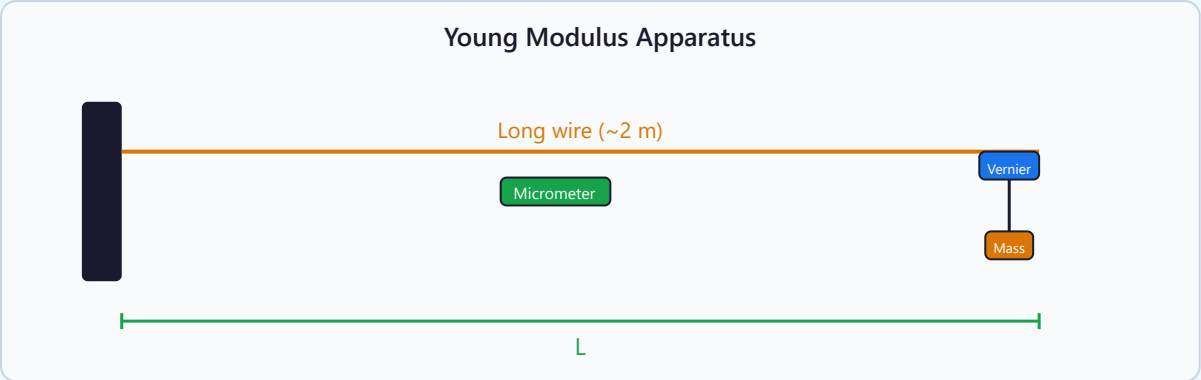
Analysis

- **Ohmic conductor:** Straight line through origin (constant resistance)
- **Filament lamp:** Curve — resistance increases as wire heats up
- **Diode:** Very little current below ~0.6 V (threshold); rapid increase above; negligible current in reverse bias

Source of Error	Type	Improvement
Component heating changes resistance	Systematic	Take readings quickly; allow cooling
Cannot reach 0 V with variable resistor alone	Systematic	Use potential divider circuit
Meter resolution	Random	Use digital meters with appropriate range
Parallax on analogue meters	Random	Read at eye level

Exam Tip: A potential divider allows you to vary the voltage from 0 V upwards, which is essential for accurately determining the diode threshold. A simple variable resistor in series cannot reduce V to 0 V.

⚗ Young Modulus of a Wire



Apparatus

- Long thin wire (~2 m), G-clamp, wooden blocks
- Micrometer, vernier scale or travelling microscope
- Slotted masses, tape measure, safety goggles

Method

1. Measure diameter (6+ readings). Calculate $A = \pi d^2/4$.
2. Measure original length L. Apply small initial load to straighten.
3. Add masses in steps. Record $F = mg$ and extension ΔL.
4. Check elastic behaviour on unloading.
5. Plot stress vs strain. Gradient = Young modulus E.

Safety

- Safety goggles; cushion beneath masses; do not exceed elastic limit

Analysis

$$E = \sigma/\varepsilon = (F/A)/(\Delta L/L)$$

Source of Error	Type	Improvement
Diameter varies	Random	Multiple readings; different positions/orientations
Wire kinks	Systematic	Apply small initial load
Thermal expansion	Systematic	Use reference wire; stable temperature

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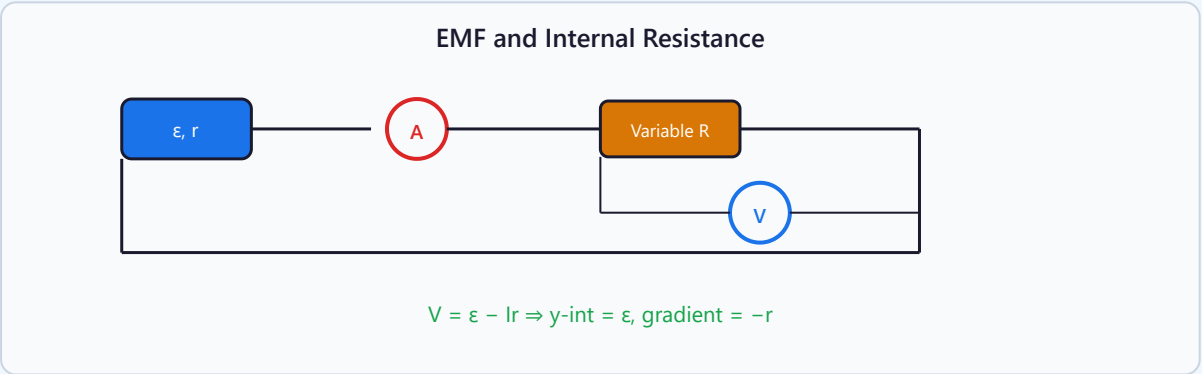
Small extensions	Random	Long thin wire; travelling microscope

Exam Tip: Percentage uncertainty in d is doubled in the area formula (d^2), making it the dominant source of error. Use a long wire to maximise extensions. The area under the linear stress-strain graph gives strain energy per unit volume.

EMF and Internal Resistance



Apparatus

- Cell, variable resistor, ammeter, voltmeter, switch

Method

1. Connect cell, ammeter, variable resistor in series. Voltmeter across cell.
2. Record V and I for 6+ settings. Switch off between.
3. Plot V vs I.

Safety

- No short circuits; switch off between readings

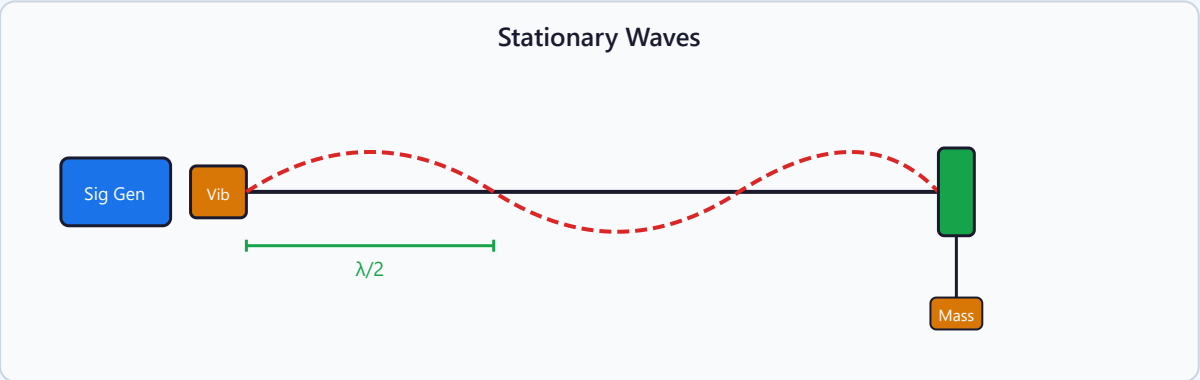
Analysis

V = ε − Ir. y-intercept = ε, gradient = −r.

Source of Error	Type	Improvement
Cell runs down	Systematic	Work quickly; switch off between readings
Meter resistances	Systematic	Digital meters
Fluctuations	Random	Wait for stable readings
Heating effects	Systematic	Short measurement bursts

Exam Tip: The x-intercept gives the short-circuit current I = ε/r. As the external resistance R increases, V approaches ε (more of the EMF is across the external component).

Stationary Waves on a String



Apparatus

- Signal generator, vibrator, string, pulley, masses, ruler

Method

1. Set up string between vibrator and pulley with hanging mass.
2. Vary frequency to find fundamental and harmonics. Record f and measure L.
3. Vary tension. Plot f vs 1/L or f vs √T.

Safety

- Safety screen; soft pad beneath masses

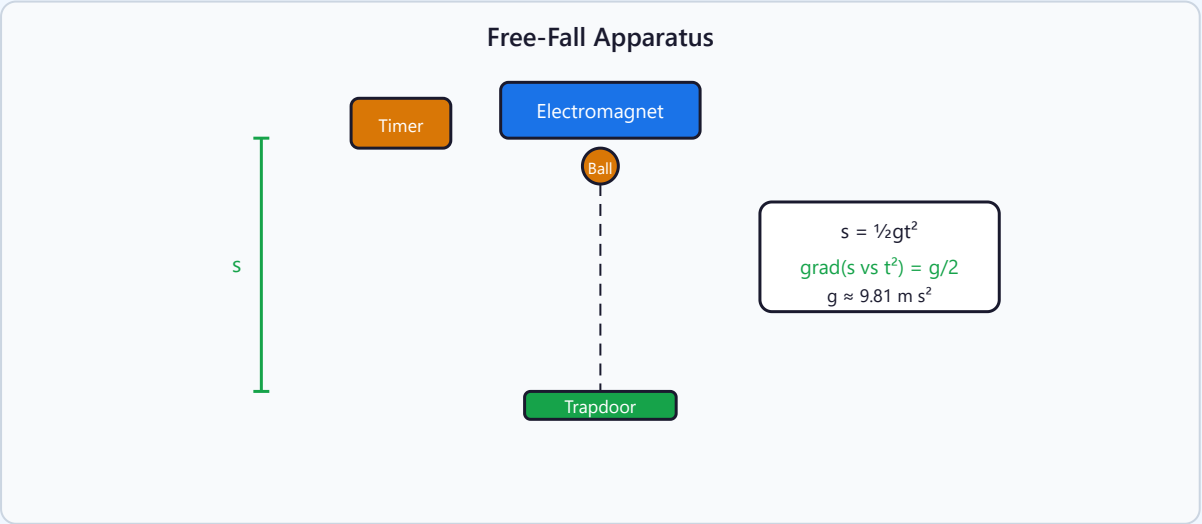
Analysis

$$f = (1/2L)\sqrt{(T/\mu)}$$

Source of Error	Type	Improvement
Resonance hard to judge	Random	Paper rider method
Length measurement	Random	Measure node to node
Hanger mass omitted	Systematic	Include in tension
String not uniform μ	Random	Weigh measured length

Exam Tip: The fundamental frequency has one antinode. Each harmonic adds one more antinode. The frequency of the nth harmonic is n times the fundamental frequency.

Determination of g by Free-Fall



Apparatus

- Electromagnet, steel ball, trapdoor, timer, 2 m ruler, clamp stand

Method

1. Set height s. Release ball (timer starts). Ball hits trapdoor (timer stops).
2. Repeat 3 times; take mean t. Vary s over 6+ values.
3. Plot s vs t^2. Gradient = g/2.

Safety

- Soft pad; stable stand; clear feet

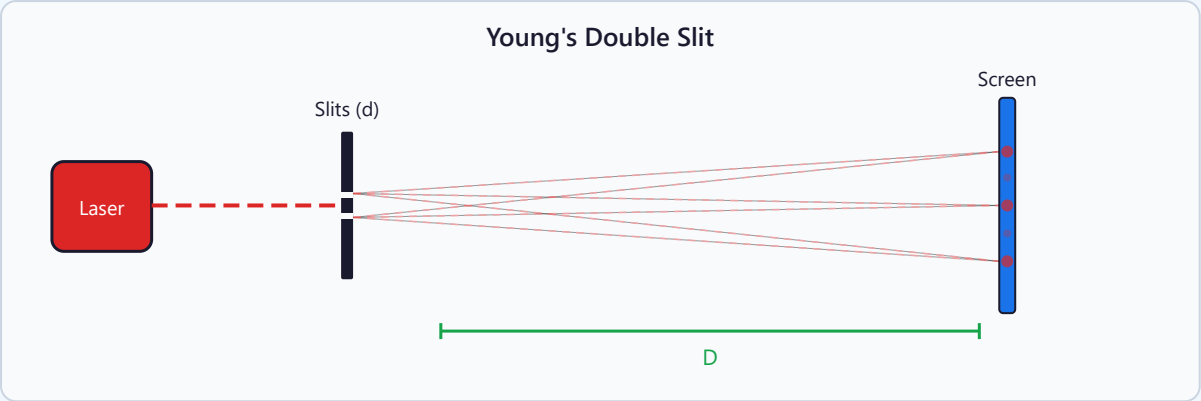
Analysis

$$s = \tfrac{1}{2}gt^2 \Rightarrow g = 2 \times \text{gradient}$$

Source of Error	Type	Improvement
Residual magnetism	Systematic	Demagnetiser or mechanical release
Air resistance	Systematic	Small dense ball
Height measurement	Random	Measure bottom of ball to trapdoor
Reaction time (if manual)	Random	Use electronic timer

Exam Tip: A systematic timing delay shifts all t values by a constant amount, giving a non-zero y-intercept on the s vs t^2 graph. The gradient is still valid. Calculate percentage error from accepted value to evaluate your result.

Investigating Interference (Double Slit)



Apparatus

- Laser (or white light + colour filter + single slit)
- Double slit (known separation d)
- Screen, ruler/vernier callipers, tape measure

Method

1. Darken room. Direct laser through double slit onto distant screen.
2. Measure across several fringe spacings. Divide to get w (fringe spacing).
3. Measure slit-to-screen distance D.
4. Calculate λ = wd/D.
5. Vary D and check λ remains consistent.

Safety

- Never look into laser beam; "Laser in use" warning
- Remove reflective objects

Analysis

$$\lambda = \mathrm{wd}/\mathrm{D}$$

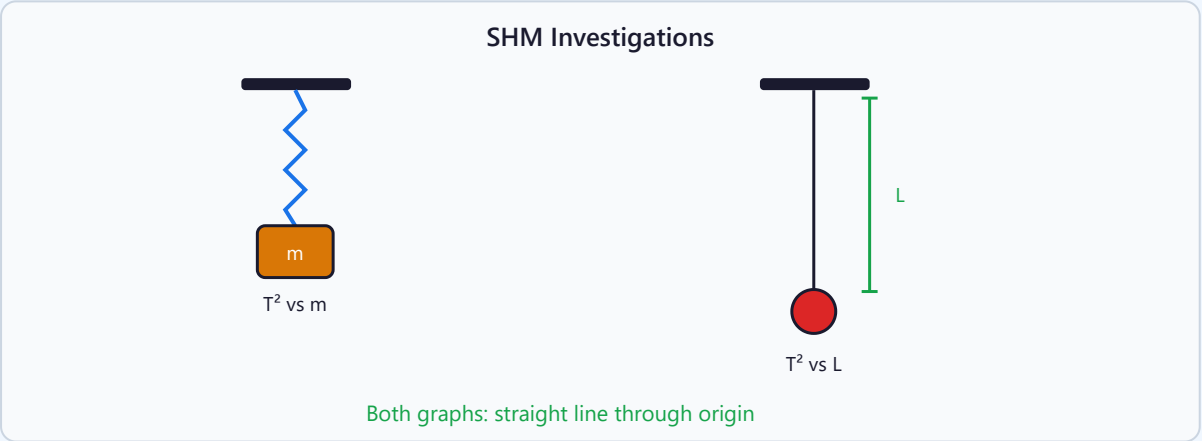
Measure across multiple fringes to reduce uncertainty in w.

Source of Error	Type	Improvement
Fringes blurred	Random	Use laser for coherent source; darken room

D measurement	Random	Large D reduces % uncertainty
Fringe spacing hard to measure	Random	Measure across 10+ fringes and divide
Laser not perpendicular to slits	Systematic	Align carefully

Exam Tip: For white light, central fringe is white with coloured fringes either side (red outermost because it has the longest wavelength and hence largest fringe spacing). With a laser, all fringes are the same colour and equally spaced.

SHM with Mass-Spring and Pendulum



Apparatus

- Spring, masses, clamp stand, stopwatch, ruler
- String, bob, protractor, ruler

Method (Mass-Spring)

1. Hang mass m. Displace slightly, release. Time 10 oscillations.
2. Repeat for 6+ masses. Plot T^2 vs m.

Method (Pendulum)

1. Measure L. Displace <10°, release. Time 10 oscillations.
2. Repeat for 6+ lengths. Plot T^2 vs L.

Safety

- Stable stand; clear of swinging masses

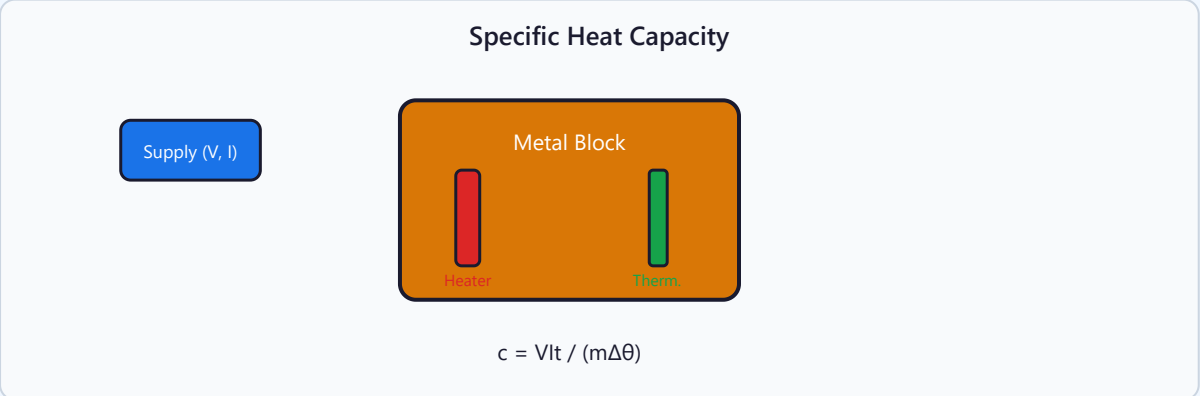
Analysis

$$\text{Spring: } T^2 = (4\pi^2/k)m \quad \text{Pendulum: } T^2 = (4\pi^2/g)L$$

Source of Error	Type	Improvement
Miscounting oscillations	Random	Fiducial marker; count from equilibrium
Large amplitude (pendulum)	Systematic	Keep θ < 10°
Reaction time	Random	Time 10+ oscillations
Spring mass	Systematic	Add $m_{spring}/3$ as effective mass

Exam Tip: The pendulum period is independent of mass. The spring period is independent of g. Plotting T^2 (not T) linearises the graph. The gradient gives k or g respectively.

Specific Heat Capacity



Apparatus

- Metal block with holes, heater, thermometer
- Ammeter, voltmeter, power supply, stopwatch, balance, insulation

Method

1. Weigh block. Insert heater and thermometer with oil.
2. Record initial θ. Heat for measured time t. Record V, I, final θ.
3. c = VIt / (mΔθ).

Safety

- Hot block; tongs/gloves; heat mat; heater fully inserted

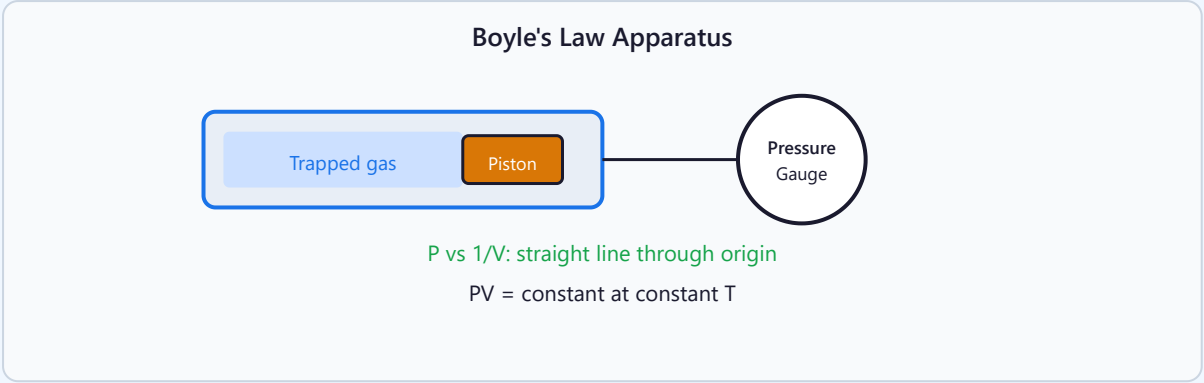
Analysis

$$c = VIt / (m\Delta\theta)$$

Source of Error	Type	Improvement
Heat loss	Systematic (c too high)	Insulate block
Poor thermal contact	Systematic	Oil in thermometer hole
V/I fluctuations	Random	Regulated supply; monitor throughout
Thermometer lag	Systematic	Wait for equilibrium; stir if liquid

Exam Tip: The measured c is always slightly too high because of heat losses. Insulation and a cooling correction can improve accuracy. Plotting Δθ vs t gives a straight line whose gradient = VI/(mc).

⚗ Boyle's Law



Apparatus

- Sealed syringe or Boyle's law apparatus, pressure gauge, thermometer

Method

1. Record initial V and P.
2. Change V slowly. Wait for temperature equilibrium.
3. Record V and P. Repeat for 6+ data points.
4. Plot P vs 1/V.

Safety

- Do not over-pressurise; safety screen

Analysis

$$\text{PV} = \text{constant} \Rightarrow \text{P vs } 1/\text{V is linear through origin}$$

Source of Error	Type	Improvement
Temperature rise on compression	Systematic	Compress slowly; wait for equilibrium
Piston friction	Systematic	Lubricate; tap gently
Gas leaks	Systematic	Check seals
Gauge parallax	Random	Read at eye level; digital gauge

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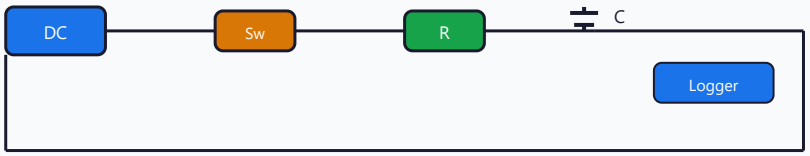
Exam Tip: Boyle's law only holds at constant temperature. If the graph of P vs 1/V does not pass through the origin, this suggests a systematic error (e.g. trapped air of unknown volume, or temperature change).

⚗ Capacitor Charge and Discharge

RC Discharge Circuit



$V = V_0 e^{(-t/RC)} \Rightarrow \ln(V)$ vs t: gradient = $-1/RC$

Apparatus

- Capacitor (470–2200 μF), resistor, DC supply, two-way switch, datalogger

Method

1. Charge capacitor through resistor. Record V vs t.
2. Discharge. Record V vs t.
3. Plot ln(V) vs t for discharge. Gradient = −1/RC.

Safety

- Correct polarity; discharge before handling; low voltage

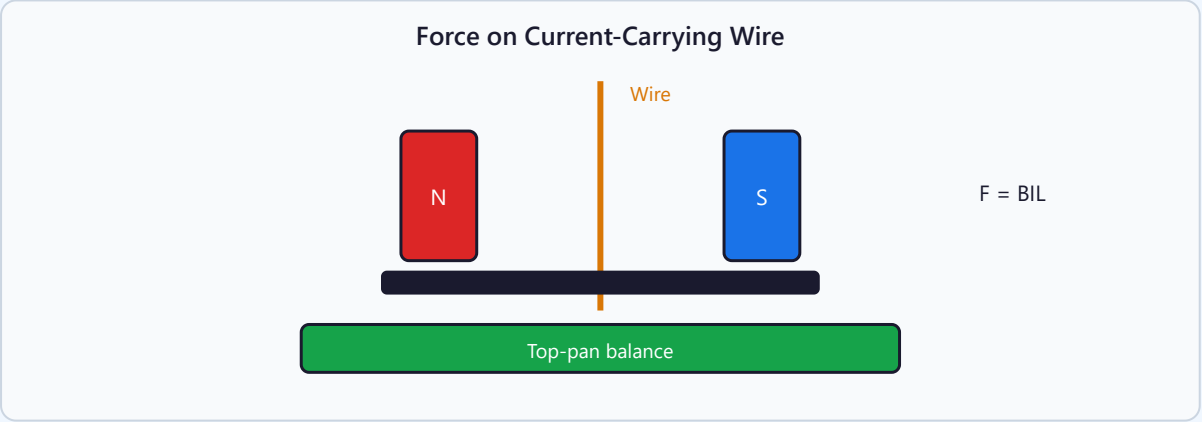
Analysis

$$V = V_0 e^{-t/RC} \Rightarrow \ln V = \ln V_0 - t/RC$$

Source of Error	Type	Improvement
Voltmeter draws current	Systematic	High-impedance datalogger
Manual timing	Random	Datalogger
Capacitor leakage	Systematic	High-quality capacitor
Component tolerance	Systematic	Measure actual R and C independently

Exam Tip: The time constant $\tau = RC$. After 1τ, V falls to 37% of V_0. After 5τ, $V < 1\%$ of V_0 (effectively fully discharged). Choose R and C so τ is 10–30 s for manageable timing.

Force on a Current-Carrying Wire



Apparatus

- U-magnet, wire cradle, top-pan balance, DC supply, ammeter, ruler

Method

1. Magnet on balance. Zero. Position wire in field.
2. Vary I. Record I and Δm. $F = \Delta m \times g$.
3. Measure wire length L in field. Plot F vs I.

4. Gradient = BL, so B = gradient/L.

Safety

- Low current; short bursts; magnets away from electronics

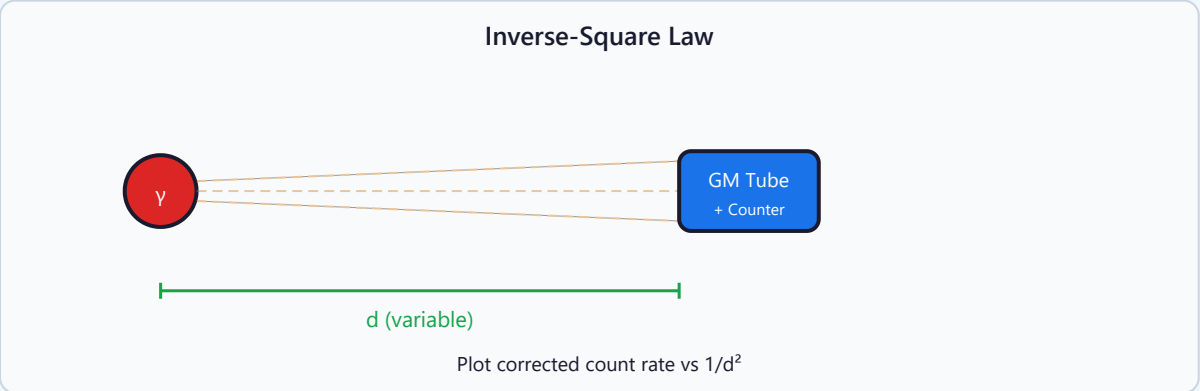
Analysis

$$F = BIL \Rightarrow \text{gradient of F vs I} = BL$$

Source of Error	Type	Improvement
Draughts/vibrations	Random	Draught shield; stable surface
Non-uniform field	Systematic	Strong flat-faced magnets; wire at centre
Wire not perpendicular	Systematic	Careful alignment
Wire heating	Systematic	Short bursts of current

Exam Tip: The balance reading changes because by Newton's third law, the force on the wire is equal and opposite to the force on the magnet. F = BIL only when B, I, and L are mutually perpendicular.

Inverse-Square Law for Gamma Radiation



Apparatus

- Gamma source, GM tube + counter, metre ruler, source holder

Method

1. Measure background (60+ s, no source).
2. Place source at distance d. Count for 60 s. Subtract background.
3. Repeat for 6+ distances. Take 3 readings at each.
4. Plot corrected count rate C vs $1/d^2$.

Safety

- Tongs; maximise distance; minimise time; ALARA; lead-lined storage

Analysis

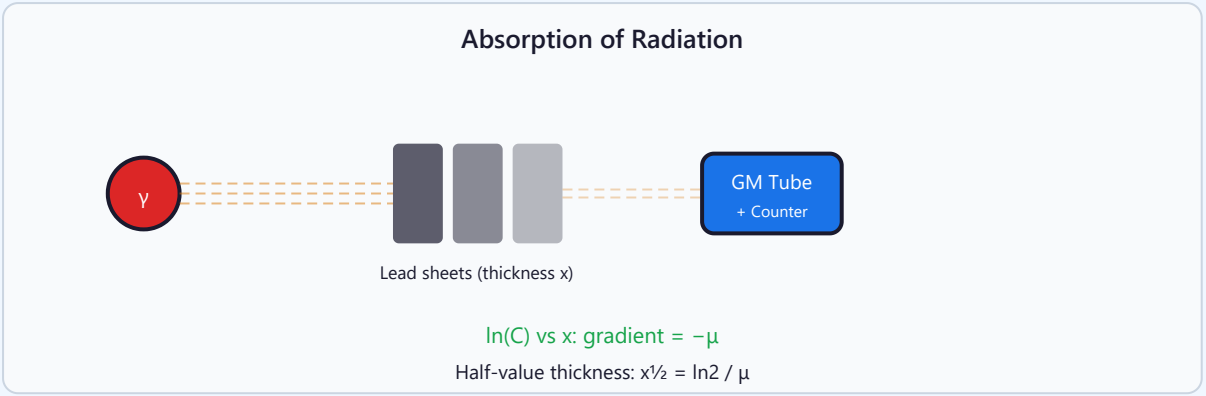
$$I \propto 1/d^2 \Rightarrow C \text{ vs } 1/d^2 \text{ should be linear through origin}$$

Or plot ln(C) vs ln(d): gradient should be −2.

Source of Error	Type	Improvement
Random decay	Random	Count for longer; more repeats
Background fluctuations	Random	Long background count
Distance to tube centre	Systematic	Measure to GM tube window
Non-point source	Systematic	Use large d values

Exam Tip: Always subtract background before plotting. If the graph curves at small d, the point-source approximation breaks down. The inverse-square law applies because gamma radiation spreads out equally in all directions from a point source.

Absorption of Radiation



Apparatus

- Gamma source, GM tube + counter, lead sheets of known thickness, callipers

Method

1. Measure background count rate (no source, 60+ s).
2. Fix source-detector distance. Record count rate with no absorber.
3. Insert lead sheets of increasing total thickness. Record corrected count rate.
4. Repeat for 6+ thicknesses.
5. Plot ln(corrected count rate) vs thickness x.

Safety

- Tongs; distance; time minimised; ALARA; lead storage

Analysis

$$I = I_0 e^{-\mu x} \Rightarrow \ln(C) \text{ vs } x \text{ gives gradient} = -\mu$$

Half-value thickness: $x_{1/2} = \ln 2 / \mu \approx 0.693 / \mu$.

Source of Error	Type	Improvement
Random nature of decay	Random	Long count times; multiple readings
Background fluctuations	Random	Long background count; multiple measurements

Absorber thickness not uniform	Systematic	Measure at multiple points with callipers
Scattered radiation	Systematic	Collimate the beam

Exam Tip: Different types of radiation have very different absorption characteristics. Alpha is stopped by paper, beta by a few mm of aluminium, and gamma requires several cm of lead. The half-value thickness is specific to both the radiation energy and the absorber material.